



Chapter 5: Comparison of Several Mean vectors (Part 1)

Zhixiang ZHANG

University of Macau

zhixzhang@um.edu.mo

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Introduction

- The ideas developed in Chapter 4 can be extended to handle problems involving the comparison of several mean vectors.
- The theories will rest on assumptions of multivariate normal distributions or large sample sizes. *Large S. S. C.L.T. MVN*
 - Small sample case:
MVN assumptions: usually F distribution
Not MVN: not considered
 - Large sample case: usually chi-squared distribution

Introduction

Lecture 4: $\mu = \mu_1 - \mu_2 = 0$

Lecture 5: $\mu_1 = \mu_2$ part 1

- We begin by considering pairs of mean vectors. Measurements are often recorded under different sets of experimental conditions to see whether the responses differ significantly over these sets. For example, the efficacy of a new drug or of an advertising campaign may be determined by comparing measurements before the “treatment” ^{new strategy} (drug or advertising) with those after the treatment.
- ^{medicine} Later we discuss comparisons among several mean vectors $H_0: \vec{\mu}_1 = \vec{\mu}_2 = \dots = \vec{\mu}_m$ ^{part 2} arranged according to treatment levels. The corresponding test statistics depend upon a partitioning of the total variation into pieces of variation attributable to the treatment sources and error, known as MANOVA.

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Paired Comparison

Let $\mathbf{x}_{1i}$ and $\mathbf{x}_{2i}$ be two p -variate responses for observation $i (i = 1, \dots, n)$.

Example: SAT pre-class test grades and post-class grades

Pre-class grades:

$$\mathbf{x}_{1i} = (\text{Quant} = 640, \text{Analyt} = 610, \text{Verbal} = 490)$$

Post-class grades:

$$\mathbf{x}_{2i} = (\text{Quant} = 680, \text{Analyt} = 620, \text{Verbal} = 560)$$

Does the class has an effect on SAT score?

Paired Comparison

1. Calculate $\mathbf{d}_i = \mathbf{x}_{1i} - \mathbf{x}_{2i}, i = 1, 2, \dots, n$. *care about difference* Denote $\delta = E[\mathbf{d}_i]$, testing $H_0: \delta = \mathbf{0}$

Assume $\mathbf{d}_i \stackrel{\text{i.i.d}}{\sim} N(\mu, \Sigma)$

$$\bar{\mathbf{d}} = \frac{1}{n} \sum_{i=1}^n \mathbf{d}_i, \text{ Testing: } T^2 = n \bar{\mathbf{d}}^\top \mathbf{S}_d^{-1} \bar{\mathbf{d}} \sim \frac{(n-1)p}{n-p} F_{p, n-p} \text{ if } H_0 \text{ is true}$$

$$\mathbf{S}_d = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{d}_i - \bar{\mathbf{d}}) (\mathbf{d}_i - \bar{\mathbf{d}})^\top$$

2. Under the assumptions that $\mathbf{d}_1, \dots, \mathbf{d}_n$ are independent samples from $N(\delta, \Sigma)$, then $T^2 = n(\bar{\mathbf{d}} - \delta)^\top \mathbf{S}_d^{-1} (\bar{\mathbf{d}} - \delta) \sim T_{p, n-1}^2 \sim \frac{(n-1)p}{n-p} F_{p, n-p}$

Same follow-up analyses as in one-sample test/intervals/regions apply here. For large sample inference, we can use χ_p^2 .

¹Here and below, we use the notation $T_{p, \nu}^2$ to represent a variable with distribution the same as $N_p(\mathbf{0}, \Sigma)^\top \left[\frac{1}{\nu} \mathbf{W}_{p, \nu}(\Sigma) \right]^{-1} N_p(\mathbf{0}, \Sigma)$

$H_0: S = 0$ $\begin{cases} T^2 > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \text{ reject, Gaussian Case} \\ T^2 > \chi^2_p(\alpha) \text{ reject if } n \text{ is large, large sample case} \end{cases}$

$H_0: a^T S = 0$ $\left| \frac{a^T \bar{d}}{\sqrt{a^T S a}} \right| > t_{n-1}(\frac{\alpha}{2})$ Gaussian Case

$\left| \frac{a^T \bar{d}}{\sqrt{a^T S a}} \right| > Z(\frac{\alpha}{2})$ quantile of large sample case standard normal

$H_0: a_1^T S = 0, \dots, a_n^T S = 0$

$\frac{a_1^T \bar{d}}{\sqrt{a_1^T S a_1}} > t_{n-1}(\frac{\alpha}{2}), \dots, \frac{a_n^T \bar{d}}{\sqrt{a_n^T S a_n}} > t_{n-1}(\frac{\alpha}{2})$ Gaussian Case when $\sum_{i=1}^n a_i = \alpha$ reject when at least one of them happen.

$\frac{a_1^T \bar{d}}{\sqrt{a_1^T S a_1}} > Z(\frac{\alpha}{2}), \dots, \frac{a_n^T \bar{d}}{\sqrt{a_n^T S a_n}} > Z(\frac{\alpha}{2})$ quantile of sd normal / reject when at least one of them happen.

currently $d_i = \chi_{1i} - \chi_{2i} \quad i=1, 2, \dots, n$
 $S = E(d_i) = 0$
 or $H_0: \mu_1 = \mu_2$

Matched pair design

$d_i = x_{1i} - x_{2i}$ on assumption of i.i.d.

but in reality this is not reasonable because it is sometimes related to other facts.

In practice, an appropriate pairing of units and a randomized assignment of treatments can enhance the statistical analysis.

like HDX's D to flip the coin of average out accumulation

Experimental Design for Paired Comparisons

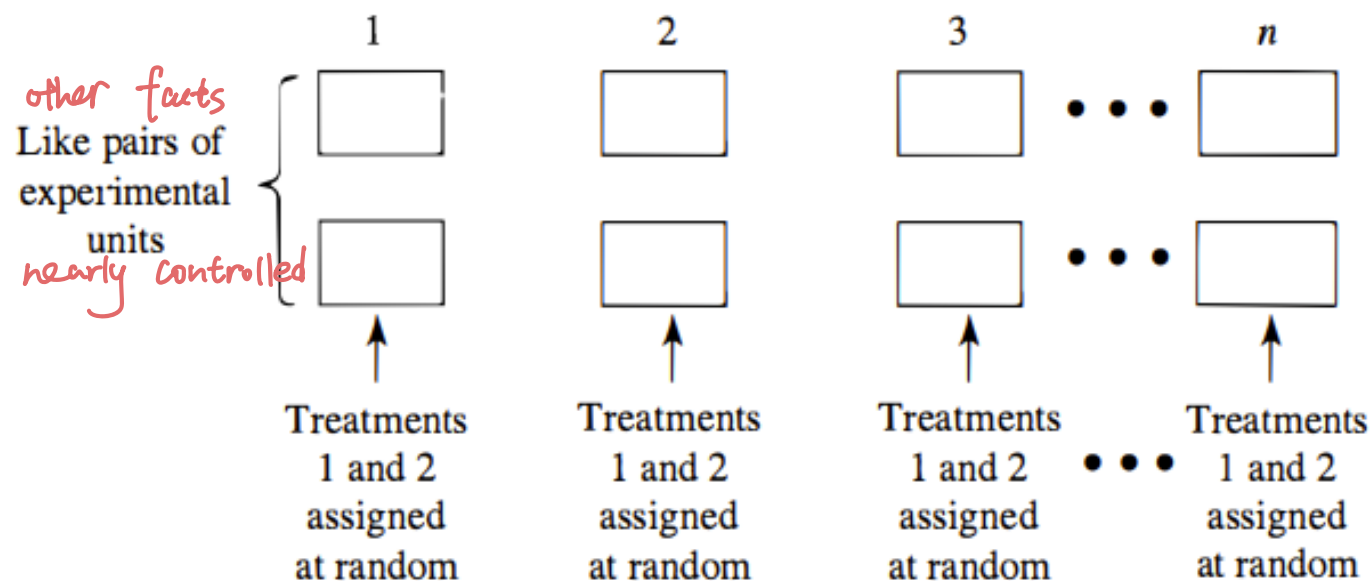


Figure: A random assignment of treatment 1 to one unit and treatment 2 to the other unit will help eliminate the systematic effects of uncontrolled sources of variation. Randomization can be implemented by flipping a coin. A separate independent randomization is conducted for each pair.

$p(\text{choosing Treatment}_i) = 0.5$ for $i=1,2$

Matched pair design

We can also think of each observation

$$\mathbf{x}_i = \begin{bmatrix} \mathbf{x}_{1i} \\ \mathbf{x}_{2i} \end{bmatrix}$$

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{\mathbf{x}}_1 \\ \bar{\mathbf{x}}_2 \end{bmatrix} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix} \quad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

Interest is in $\mathbf{C}\mathbf{x}_i$, where

$$H_0: E(x_{1i} - x_{2i}) = 0$$

name: contrast matrix

$$\mathbf{C} = \begin{bmatrix} \overset{(1)}{1} & 0 & \dots & \overset{(p)}{0} & \overset{(p+1)}{-1} & 0 & \dots & \overset{(2p)}{0} \\ 0 & 1 & \dots & 0 & 0 & -1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 & 0 & \dots & -1 \end{bmatrix} \begin{bmatrix} x_{11} \\ \vdots \\ x_{1p} \\ x_{21} \\ \vdots \\ x_{2p} \end{bmatrix} = \begin{bmatrix} x_{11} - x_{21} \\ \vdots \\ x_{1p-1} - x_{2p-1} \\ x_{1p} - x_{2p} \end{bmatrix}$$

(2p x 1)

Matched pair design comparison

Note that

$$\text{Denote: } \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$C\mu = \mu_1 - \mu_2 \in \mathbb{R}^{p \times 1}$$

$$\mathbf{d}_i = C\mathbf{x}_i$$

$$\bar{\mathbf{d}} = C\bar{\mathbf{x}}$$

$$\mathbf{S}_d = CSC^T$$

Under the assumption that $\mathbf{d}_i$ are independent $N(0, \Sigma)$

$$T^2 = n\bar{\mathbf{x}}^T C^T (CSC^T)^{-1} C\bar{\mathbf{x}} \sim T_{p, n-1}^2$$

$$\sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

$$N(\mu, \Sigma)^T \left(\frac{1}{n-1} W(n-1, \Sigma) \right)^{-1} N(\mu, \Sigma)$$

$$\sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

$H_0: \mu_1 = \mu_2 = \dots = \mu_K$

Let $C = \begin{pmatrix} 1 & -1 & & \\ & 1 & -1 & \\ & & \ddots & \ddots \\ & & & 1 & -1 \end{pmatrix}$, $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_K \end{pmatrix}$ So $\mu_1 = \dots = \mu_K \Leftrightarrow C\mu = 0$

because $C\mu = \begin{pmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \\ \vdots \\ \mu_{K-1} - \mu_K \end{pmatrix} = 0 \Leftrightarrow \begin{cases} \mu_1 = \mu_2 \\ \mu_2 = \mu_3 \\ \vdots \\ \mu_{K-1} = \mu_K \end{cases} \Leftrightarrow \mu_1 = \dots = \mu_K$

because $C\bar{\mathbf{x}}$ is approximation $C\mu$ and $E[\bar{\mathbf{x}}] = \mu \Rightarrow E[C\bar{\mathbf{x}}] = CE[\bar{\mathbf{x}}] = C\mu$

If $X_i \stackrel{i.i.d.}{\sim} N(\mu, \Sigma)$, for any $C \in \mathbb{R}^{m \times p}$, $m < p$, the Hotelling's T^2 based on $\{C\mathbf{x}_1, C\mathbf{x}_2, \dots, C\mathbf{x}_n\}$ which is $n[C(\bar{\mathbf{x}} - \mu)]^T [CSC^T]^{-1} [C(\bar{\mathbf{x}} - \mu)]$

$\sim \frac{(n-1)m}{n-m} F_{m, n-m}$ dist. doesn't depend on C

$\sim T_{m, n-1}^2$ here $m = K-1$ because $C \in \mathbb{R}^{(K-1) \times K}$ if C invertible

$n(\bar{\mathbf{x}} - \mu)^T C^T (C^T)^{-1} S^{-1} C^{-1} C(\bar{\mathbf{x}} - \mu) = n(\bar{\mathbf{x}} - \mu)^T S^{-1} (\bar{\mathbf{x}} - \mu)$ no dependence on C

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Setting: Group 1: $X_{11}, X_{12}, \dots, X_{1n_1}$ n_1 observations, each has P features μ_1, Σ_1
 Group 2: $X_{21}, X_{22}, \dots, X_{2n_2}$ n_2 observations, each has P features. μ_2, Σ_2
 $H_0: \mu_1 = \mu_2$

1 Paired Comparison

2 Comparing means from two populations

3 One-Way MANOVA

Comparing means from two populations

Interest in $\mu_1 - \mu_2$ (difference in two population means). Generalizing Hotelling's T^2 to two populations.

Assumptions:

- $\mathbf{x}_{11}, \mathbf{x}_{12}, \dots, \mathbf{x}_{1n_1}$ is a random sample of size n_1 from a p -variate population with mean vector μ_1 and covariance matrix Σ_1 .
- $\mathbf{x}_{21}, \mathbf{x}_{22}, \dots, \mathbf{x}_{2n_2}$ is a random sample of size n_2 from a p -variate population with mean vector μ_2 and covariance matrix Σ_2 .²
- The two samples are independent

Let

$$\bar{\mathbf{x}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{x}_{ij}, \quad i = 1, 2$$

$$\mathbf{S}_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i) (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)^\top, \quad i = 1, 2$$

²Note that two groups do not need have the same number of samples

In addition: $\Sigma_1 = \Sigma_2$, X_{1i} , X_{2j} MVN

If in addition assume $\Sigma_1 = \Sigma_2 = \Sigma$ and two samples are MVN, then

$$E[\bar{X}_1 - \bar{X}_2] = E[\bar{X}_1] - E[\bar{X}_2] = \mu_1 - \mu_2$$

$$(\bar{X}_1 - \bar{X}_2) \sim N_p \left(\mu_1 - \mu_2, \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \Sigma \right).$$

Note that

$$(n_1 - 1) \mathbf{S}_1 \sim W_p(n_1 - 1, \Sigma)$$

$$(n_2 - 1) \mathbf{S}_2 \sim W_p(n_2 - 1, \Sigma)$$

$$= N_p(0, \frac{\Sigma}{n_1} + \frac{\Sigma}{n_2} - 0 - 0)$$

$$= N_p(0, (\frac{1}{n_1} + \frac{1}{n_2}) \Sigma)$$

and

$$\underbrace{(n_1 - 1) \mathbf{S}_1 + (n_2 - 1) \mathbf{S}_2}_{=(n_1 + n_2 - 2) \mathbf{S}_{pooled}} \sim W_p(n_1 + n_2 - 2, \Sigma).$$

Define

$$\mathbf{S}_{pooled} = \frac{\sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1) (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)^\top + \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2) (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)^\top}{n_1 + n_2 - 2}$$

$$= \frac{n_1 - 1}{n_1 + n_2 - 2} \mathbf{S}_1 + \frac{n_2 - 1}{n_1 + n_2 - 2} \mathbf{S}_2$$

then

$$E[\mathbf{S}_{pooled}] = \Sigma$$

Test $H_0: \mu_1 = \mu_2 \in \mathbb{R}^p$

Population 1 $X_{11}, X_{12}, \dots, X_{1n_1}$ $X_{1i} \sim N_p(\mu_1, \Sigma_1)$

Population 2 $X_{21}, X_{22}, \dots, X_{2n_2}$ $X_{2i} \sim N_p(\mu_2, \Sigma_2)$

If $\Sigma_1 = \Sigma_2 (= \Sigma)$, then

$$\bar{X}_1 - \bar{X}_2 \sim N_p(\mu_1 - \mu_2, (\frac{1}{n_1} + \frac{1}{n_2}) \Sigma)$$

How to estimate Σ ?

$$S_1 = \frac{1}{n_1-1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)(X_{1i} - \bar{X}_1)^T \quad E[S_1] = \Sigma$$

$$S_2 = \frac{1}{n_2-1} \sum_{i=1}^{n_2} (X_{2i} - \bar{X}_2)(X_{2i} - \bar{X}_2)^T \quad E[S_2] = \Sigma$$

a better estimate:

$$S_{\text{pooled}} = \frac{n_1-1}{n_1+n_2-2} S_1 + \frac{n_2-1}{n_1+n_2-2} S_2, \quad E[S_{\text{pooled}}] = \Sigma$$

Use S_{pooled} to estimate Σ

$$[(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)]^T \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \Sigma \right]^{-1} [(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)] \sim \chi_p^2$$

by: if $Z \sim N_p(0, \Sigma)$ independent of $V \sim W_p(n-1, \Sigma) \sim \frac{1}{p} T^2_{p, n-1, 2}$

$$\text{then } Z^T \left(\frac{V}{n-1} \right)^{-1} Z \sim T_{p, n-1}$$

$$\sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

$$(n_1+n_2-2) S_{\text{pooled}}$$

$$= (n_1-1) S_1 + (n_2-1) S_2$$

$$= W_p(n_1-1, \Sigma) + W_p(n_2-1, \Sigma) \sim W_p(n_1+n_2-2, \Sigma)$$

independent

D. Test:

So reject H_0 if

$$(\bar{X}_1 - \bar{X}_2)^T \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{\text{pooled}} \right]^{-1} (\bar{X}_1 - \bar{X}_2) > \frac{(n_1+n_2-2)p}{n_1+n_2-p-1} F_{p, n_1+n_2-p-1}(\alpha)$$

$$\text{Cov}(\bar{X}_1 - \bar{X}_2) = \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \Sigma$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0, \Sigma)$$

$$W_1 = \sum_{i=1}^{d_1} Z_i Z_i^T, \quad Z_i \in N_p(0, \Sigma)$$

$$W_2 = \sum_{i=1}^{d_2} \tilde{Z}_i \tilde{Z}_i^T, \quad \tilde{Z}_i \in N_p(0, \Sigma)$$

$$\text{then } W_1 + W_2 \sim W_p(d_1+d_2, \Sigma)$$

$$\text{We may could } S_{\text{new}} := c_1 S_1 + c_2 S_2$$

$$c_1 + c_2 = 1$$

$$E[S_{\text{new}}] = \Sigma$$

$$\Delta \quad W_1 = \sum_{i=1}^{d_1} Z_i Z_i^T, \quad Z_i \sim N_p(0, \Sigma)$$

$$W_2 = \sum_{i=1}^{d_2} \tilde{Z}_i \tilde{Z}_i^T, \quad \tilde{Z}_i \sim N_p(0, \Sigma)$$

$$\text{then } W_1 + W_2 \sim W_p(d_1+d_2, \Sigma)$$

$$\text{Var} \left[c_1 \frac{1}{n_1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2 + c_2 \frac{1}{n_2} \sum_{i=1}^{n_2} (X_{2i} - \bar{X}_2)^2 \right]$$

$$\text{Recall: if } Z \sim N_p(0, \Sigma) \text{ indep then } Z^T \left(\frac{V}{n-1} \right)^{-1} Z \sim T_{p, n-1}$$

$$c_1 = \frac{n_1}{n_1+n_2}, \quad c_2 = \frac{n_2}{n_1+n_2}$$

$$S_{\text{new}} = \frac{c_1}{c_1+c_2} S_1 + \frac{c_2}{c_1+c_2} S_2$$

$$E[S_{\text{new}}] = \Sigma$$

$$\text{Sample mean: } \bar{X}_i = \frac{n_i-1}{n_1+n_2-2}$$

When $\Sigma_1 = \Sigma_2 = \Sigma$ and two samples are MVN

Recall that

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) \sim N_p \left(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \Sigma \right)$$

then

$$\begin{aligned} T^2 &= [\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{pooled} \right]^{-1} [\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \\ &\sim T_{p, n_1 + n_2 - 2}^2 \\ &\sim \frac{(n_1 + n_2 - 2) p}{(n_1 + n_2 - 2) - p + 1} F_{p, (n_1 + n_2 - 2) - p + 1} \end{aligned}$$

Follow-up analyses

- “t-interval”:

$$\mathbf{a}^\top \bar{\mathbf{x}}_1 - \mathbf{a}^\top \bar{\mathbf{x}}_2 \pm t_{n_1+n_2-2} \left(\frac{\alpha}{2} \right) \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{a}^\top \mathbf{S}_{pooled} \mathbf{a}}$$

- “Bonferroni interval”:

$$\mathbf{a}_i^\top \bar{\mathbf{x}}_1 - \mathbf{a}_i^\top \bar{\mathbf{x}}_2 \pm t_{n_1+n_2-2} \left(\frac{\alpha}{2k} \right) \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{a}_i^\top \mathbf{S}_{pooled} \mathbf{a}_i}$$

for $i = 1, \dots, k$;

- “ T^2 -interval”

$$\mathbf{a}^\top \bar{\mathbf{x}}_1 - \mathbf{a}^\top \bar{\mathbf{x}}_2 \pm \sqrt{T_{\alpha, p, n_1+n_2-2}^2} \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{a}^\top \mathbf{S}_{pooled} \mathbf{a}}$$

where $T_{\alpha, p, n_1+n_2-2}^2 \equiv \frac{(n_1+n_2-2)p}{(n_1+n_2-2)-p+1} F_{p, (n_1+n_2-2)-p+1}(\alpha)$

When $\Sigma_1 \neq \Sigma_2$

- When $\Sigma_1 \neq \Sigma_2$, only assuming MVN is not enough, so we consider the large sample case.
- As $n_1 \rightarrow \infty$, $n_2 \rightarrow \infty$, by the central limit theorem, $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$ is approximately $N_p[\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, n_1^{-1}\Sigma_1 + n_2^{-1}\Sigma_2]$, thus

$$[\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left(\frac{1}{n_1} \Sigma_1 + \frac{1}{n_2} \Sigma_2 \right)^{-1} [\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \rightarrow \chi_p^2$$

When n_1 and n_2 are large, with high probability, $\mathbf{S}_1 \rightarrow \Sigma_1$, $\mathbf{S}_2 \rightarrow \Sigma_2$. Consequently, the approximation holds with $\mathbf{S}_1$ and $\mathbf{S}_2$ in place of Σ_1 and Σ_2 , respectively; and we have

$$[\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left[\frac{\mathbf{S}_1}{n_1} + \frac{\mathbf{S}_2}{n_2} \right]^{-1} [\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \rightarrow \chi_p^2$$

- Follow-up analysis in test/confidence intervals/regions apply.

$$x \sim N(\mu, \Sigma) \quad (x - \mu)^T \Sigma^{-1} (x - \mu) \sim \chi_p^2$$

As $n \rightarrow \infty$ when $\bar{x}_1 \neq \bar{x}_2$

$$[\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)]^T \left(\frac{1}{n_1} \Sigma_1 + \frac{1}{n_2} \Sigma_2 \right)^{-1} [\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)] \rightarrow \chi_p^2$$

$$\uparrow \quad \uparrow$$

$$\frac{1}{n_1} \Sigma_1 + \frac{1}{n_2} \Sigma_2$$

$$\rightarrow \chi_p^2 \quad \checkmark$$

$$S \rightarrow \Sigma$$

no nail biting

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Comparing of several means

A medical study that aims to compare the effectiveness of several different treatment approaches (Treatment A, Treatment B, ...) for managing a particular chronic disease, such as diabetes. The study could assess various health markers, including blood sugar levels, cholesterol levels, and blood pressure.

The collected data can be organized as

		Treatment			
		1	2	...	g
Subjects	1	$\mathbf{x}_{11}$	$\mathbf{x}_{21}$	...	$\mathbf{x}_{g1}$
	2	$\mathbf{x}_{12}$	$\mathbf{x}_{22}$	...	$\mathbf{x}_{g2}$
	$\vdots$	$\vdots$	$\vdots$		$\vdots$
	n_i	$\mathbf{x}_{1n_1}$	$\mathbf{x}_{2n_2}$	...	$\mathbf{x}_{gn_g}$

where each $\mathbf{x}_{\ell j}$ is p -variate vector.

Basic Assumptions

Assumptions needed for statistical inference.

- $\mathbf{x}_{/1}, \mathbf{x}_{/2}, \dots, \mathbf{x}_{/n_l}$ is a random sample of size n_l from a population with means μ_l for $l = 1, \dots, g$ (i.e., observations within populations are independent and representative of their populations).
- Random samples from different populations are independent.
- All populations have the same covariance matrix, Σ .
- $\mathbf{x}_{/j} \sim \mathcal{N}(\mu_l, \Sigma)$; that is, each population is multivariate normal.

If a population is not multivariate normal, then for large n_l central limit theorem may "kick-in".

Aim: $H_0: \mu_1 = \mu_2 = \dots = \mu_g$

no nail biting

$X_{ij} \sim N(\mu_i, \Sigma)$

MANOVA
↓
multivariate analysis of variance.

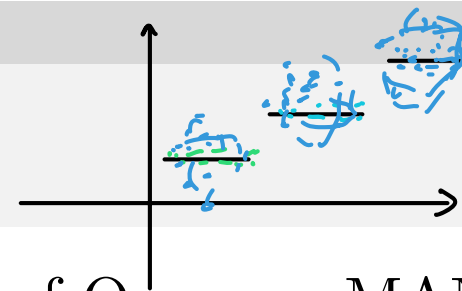
$p > 1$: each X_{ij} is a vector.

One-way ANOVA

$$\mu_b = \mu + \mu_b - \mu$$

common mean for all groups
grand mean

b-treatment effect.



We start by considering the case where $p = 1$ of One-way MANOVA, also known as One-way ANOVA

- Assumptions: $X_{lj} \sim \mathcal{N}(\mu_l, \sigma^2)$ i.i.d for $j = 1, \dots, n_l$ and $l = 1, \dots, g$.
- Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_g \quad \text{versus} \quad H_1 : \text{not } H_0$$

- We usually express μ_l as the sum of a grand mean and deviations from the grand mean

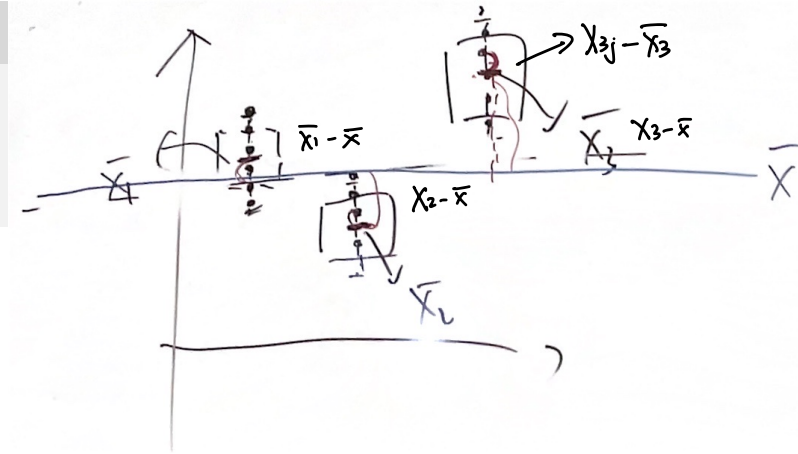
$$\underbrace{\mu_l}_{\ell^{\text{th}} \text{ pop. mean}} = \underbrace{\mu}_{\text{grand mean}} + \underbrace{\mu_l - \mu}_{\ell^{\text{th}} \text{ pop. treatment effect}}$$

$$= \mu + \tau_l$$

- If $\mu_1 = \mu_2 = \dots = \mu_g$, then an equivalent way to write the null hypothesis is

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_g = 0$$

The model for an observation



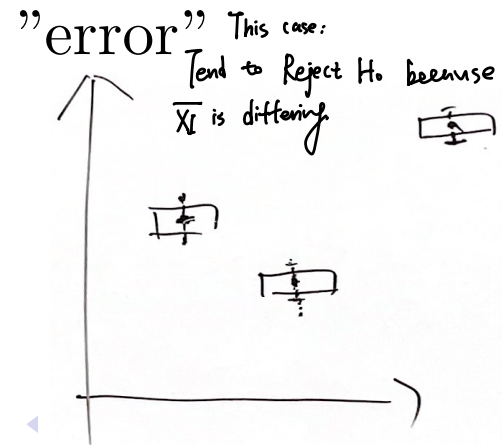
$$X_{lj} = \mu + \tau_l + \epsilon_{lj}$$

where $\epsilon_{lj} \sim \mathcal{N}(0, \sigma^2)$ and independent.

- ϵ_{lj} is "random error".
- We typically impose the condition $\sum_{l=1}^g \tau_l = 0$ as an identification constraint.
- The decomposition of an observation is no shift to μ

$$\underbrace{X_{lj}}_{\text{observation}} = \underbrace{\bar{X}}_{\text{overall sample mean}} + \underbrace{(\bar{X}_l - \bar{X})}_{\text{estimated treatment effect}} + \underbrace{(X_{lj} - \bar{X}_l)}_{\text{residual "error"}}$$

- $\bar{X}$ is the estimator of μ
- $\hat{\tau}_l = (\bar{X}_l - \bar{X})$ is the estimator of τ_l
- $(X_{lj} - \bar{X}_l)$ is the estimator of ϵ_{lj} .



Sum of squares

The total sum of squares:

$$SS_{\text{total}} = \sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X})(X_{lj} - \bar{X})^\top$$

$n = \sum_{l=1}^g n_l$

Remark: Here we use the “transpose” to indicate that the decomposition discussed in the next two slides also applies to multivariate cases, i.e., you can directly replace X_{lj} with $\mathbf{x}_{lj}$, $\bar{X}_l$ with $\bar{\mathbf{x}}_l$...

Note that

$$\begin{aligned} (X_{lj} - \bar{X})(X_{lj} - \bar{X})^\top &= [(X_{lj} - \bar{X}_l) + (\bar{X}_l - \bar{X})] [(X_{lj} - \bar{X}_l) + (\bar{X}_l - \bar{X})]^\top \\ &= \underbrace{(X_{lj} - \bar{X}_l)(X_{lj} - \bar{X}_l)^\top}_{\text{squares-products}} + \underbrace{(\bar{X}_l - \bar{X})(\bar{X}_l - \bar{X})^\top}_{\text{significant in scale}} \\ &\quad + \underbrace{\left[(X_{lj} - \bar{X}_l)(\bar{X}_l - \bar{X})^\top + (\bar{X}_l - \bar{X})(X_{lj} - \bar{X}_l)^\top \right]}_{\text{cross-products}} = 0 \end{aligned}$$

$\sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l)(X_{lj} - \bar{X}_l)^\top = SS_{\text{residual}}$
 $\sum_{l=1}^g n_l (\bar{X}_l - \bar{X})(\bar{X}_l - \bar{X})^\top = SS_{\text{treatment}}$

Sum of cross-products terms

$$\begin{aligned}
 & \sum_{l=1}^g \left(\sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l) (\bar{X}_l - \bar{X})^\top \right) = 0 \\
 & \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l) (\bar{X}_l - \bar{X})^\top = \left(\sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l) \right) (\bar{X}_l - \bar{X})^\top \\
 & = \left(\left(\sum_{j=1}^{n_l} X_{lj} \right) - n_l \bar{X}_l \right) (\bar{X}_l - \bar{X})^\top \\
 & = n_l \underbrace{(\bar{X}_l - \bar{X}_l)}_0 (\bar{X}_l - \bar{X})^\top = 0
 \end{aligned}$$

Sum of Squares decomposition

Now summing the rest over j and ℓ we get

$$\sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X}) (X_{lj} - \bar{X})^\top = \sum_{l=1}^g n_l (\bar{X}_l - \bar{X}) (\bar{X}_l - \bar{X})^\top$$

$$+ \sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l) (X_{lj} - \bar{X}_l)^\top$$

$$\begin{aligned} \text{Total SS} &= \text{Treatment} + \text{Residual} \\ &= \text{Between Groups} + \text{Within Groups} \\ &= \text{Hypothesis} + \text{Error} \\ &\quad H_0: \tau_1 = \tau_2 = \dots = \tau_I = 0 \end{aligned}$$

ANOVA table and test

Let $n_+ = \sum_{l=1}^g n_l$, the total sample size

Source of Variation	Sum of Squares	df
Treatment	$SS_{tr} = \sum_{l=1}^g n_l (\bar{X}_l - \bar{X})^2$	$g - 1$
Residual	$SS_{res} = \sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l)^2$	$n_+ - g$
Total	$= \sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X})^2$	$n_+ - 1$

Test statistic for $H_0 : \mu_1 = \cdots = \mu_g$ (or $H_0 : \tau_1 = \cdots = \tau_g$) and its sampling distribution are

$$F = \frac{SS_{tr} / (g - 1)}{SS_{res} / (n_+ - g)} \sim \mathcal{F}_{(g-1), (n_+ - g)}$$

ANOVA test

Between Groups is large and Within Groups is small

The ANOVA F -test rejects $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = 0$ at level α if

$$F = \frac{SS_{\text{tr}}/(g-1)}{SS_{\text{res}}/(\sum_{\ell=1}^g n_{\ell} - g)} > F_{g-1, \sum n_{\ell} - g}(\alpha)$$

This is equivalent to rejecting H_0 for large values of $SS_{\text{tr}}/SS_{\text{res}}$ or for large values of $1 + SS_{\text{tr}}/SS_{\text{res}}$. The statistic appropriate for a multivariate generalization rejects H_0 for small values of the reciprocal

$$\frac{1}{1 + SS_{\text{tr}}/SS_{\text{res}}} = \frac{SS_{\text{res}}}{SS_{\text{res}} + SS_{\text{tr}}}$$

A closer look at Within group SS

$$\begin{aligned}
 \mathbf{W} &= \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)^\top \\
 &= \sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1) (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)^\top + \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2) (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)^\top \\
 &\quad \cdots + \sum_{j=1}^{n_g} (\mathbf{x}_{gj} - \bar{\mathbf{x}}_g) (\mathbf{x}_{gj} - \bar{\mathbf{x}}_g)^\top \\
 &= \mathbf{W}_1 + \mathbf{W}_2 + \cdots + \mathbf{W}_g \\
 &= (n_1 - 1) \mathbf{S}_1 + (n_2 - 1) \mathbf{S}_2 + \cdots + (n_g - 1) \mathbf{S}_g
 \end{aligned}$$

$\mathbf{S}_i$ is the sample covariance matrix for the i -th group (treatment, condition, etc).

Test statistic

With respect to between groups SS,

$$\mathbf{B} = \sum_{l=1}^g n_l \overset{\hat{\tau}_l: \text{vector}}{(\bar{\mathbf{x}}_l - \bar{\mathbf{x}})} \overset{\hat{\tau}_l^T: \text{vector}}{(\bar{\mathbf{x}}_l - \bar{\mathbf{x}})}^\top = \sum_{l=1}^g n_l \overset{\text{matrix}}{\hat{\tau}_l \hat{\tau}_l^\top}$$

- If $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = \mathbf{0}$ is true, Then $\mathbf{B}$ should be "close" to $\mathbf{0}$.
each entry close to 0
- To test H_0 , we consider the ratio of generalized SSs,

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{B}|} = \frac{|\mathbf{W}|}{|\mathbf{T}|}$$

where $\mathbf{T} = \mathbf{W} + \mathbf{B}$.

- Λ^* is known as "Wilk's Lambda".
- It's equivalent to likelihood ratio ^{method} statistic.

Hypothesis testing with Λ^*

Λ^* is a ratio of generalized sampling variances

$T = W + B$
 $W \succ 0, B \succ 0$
 $\lambda_i^* := \lambda_i(T) \geq \lambda_i := \lambda_i(W)$

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{T}|} = \frac{\prod_{i=1}^p \lambda_i}{\prod_{i=1}^p \lambda_i^*} \leq 1$$

$$= \frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{B}|}$$

- where λ_i are eigenvalues of $\mathbf{W}$, and λ_i^* are eigenvalues of $\mathbf{T}$.
- If $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = 0$ is true then $\mathbf{B}$ is close to 0

$$\Rightarrow \mathbf{T} \approx \mathbf{W} \Rightarrow \lambda_i \approx \lambda_i^* \Rightarrow \Lambda^* \text{ close to } 1.$$

- If $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = \mathbf{0}$ is false then $\mathbf{B}$ is not close 0
 $\Rightarrow \lambda_i < \lambda_i^* \Rightarrow \Lambda^*$ is "small".

The exact distribution of Λ^* can be derived for special cases of p and g .

Distribution of Wilks's Lambda Λ^* B
w

No. of variables	No. of groups	Sampling distribution
$p = 1$	$g \geq 2$	$\left(\frac{\sum n_\ell - g}{g-1} \right) \left(\frac{1 - \Lambda^*}{\Lambda^*} \right) \sim F_{g-1, \sum n_\ell - g}$
$p = 2$	$g \geq 2$	$\left(\frac{\sum n_\ell - g - 1}{g-1} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) \sim F_{2(g-1), 2(\sum n_\ell - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\frac{\sum n_\ell - p - 1}{p} \right) \left(\frac{1 - \Lambda^*}{\Lambda^*} \right) \sim F_{p, \sum n_\ell - p - 1}$
$p \geq 1$	$g = 3$	$\left(\frac{\sum n_\ell - p - 2}{p} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) \sim F_{2p, 2(\sum n_\ell - p - 2)}$

If H_0 is true and $\sum n_\ell = n$ is large,

$$- \left(n - 1 - \frac{(p + g)}{2} \right) \ln \Lambda^*$$

has approximately a chi-square distribution with $p(g - 1)$ d.f.

$$T = W + B$$

$$W \geq 0, B \geq 0$$

$$\lambda_i^* = \lambda_i(T) \geq \lambda_i(W)$$

$$\lambda^* = \frac{|W|}{|T|} = \frac{\frac{p}{\prod_{i=1}^p \lambda_i} \lambda_i y}{\frac{p}{\prod_{i=1}^p \lambda_i}}$$

$$(1 \ 0 \dots 0)$$

$$n = \sum n_l$$

$$SS_{tot} = \sum_{l=1}^g \sum_{i=1}^{n_l} (x_{lj} - \bar{x})(x_{lj} - \bar{x})^T$$

$$x_{lj} = \bar{x} + \bar{x}_l - \bar{x} + x_{lj} - \bar{x}$$

$$\sum_{l=1}^g \sum_{i=1}^{n_l} (\bar{x}_l - \bar{x})(\bar{x}_l - \bar{x})^T$$

$$X = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

$$y_l = \mu + \beta_1 x_{l1} + \beta_2 x_{l2} + \dots + \beta_g x_{lg}$$

If y_{lj} is in group l_0
then $x_{l0} = 1$
and $x_{lj} = 0$
if $j \neq l_0$

$$\sum_{l=1}^g \left(\sum_{j=1}^g (x_{lj} - \bar{x}_l) \right) (\bar{x}_l - \bar{x})^T = 0$$

Summary of One-way MANOVA

MANOVA table:

Source of variation	Sum of Squares	df	Λ^*
Treatment (Between)	$\mathbf{B} = \sum_{l=1}^g n_l (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})^\top$	$g - 1$	$ \mathbf{W} / \mathbf{T} $
Residual (Within)	$\mathbf{W} = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)^\top$	$n - g$	
Total (corrected for mean)	$\mathbf{T} = \mathbf{W} + \mathbf{B}$ $= \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}) (\mathbf{x}_{lj} - \bar{\mathbf{x}})^\top$	$n - 1$	

Reject H_0 when the above-mentioned statistics involving Λ^* is greater than the associated critical values.

Inference on several means

Compare means for two populations μ_1, μ_2

more than two population MANOVA

1-way
MANOVA.

$\downarrow$
 $\mu + \tau_i$

two-way
MANOVA

interaction term

$\downarrow$
test for interaction effect

$\mu + \tau_A + \tau_B + \tau_{AB}$